

Bill: H. R. 9510 (illustrative; the Clerk assigns the number at introduction), 119th Congress, 2d Session. **Refer to:** House Energy and Commerce (Health)

One-Page Summary

Prepared by Kevin Kawchak , CEO ChemicalQDevice | kevink@chemicalqdevice.com | [10.5281/zenodo.xxxxxxxx](https://doi.org/10.5281/zenodo.xxxxxxxx)

Independent research draft. Not an enacted law and not legal advice; not endorsed by the FDA, HHS, or any Member of Congress.

Subject

The problem

A sequencing gap that embodiment makes dangerous: Physical AI oncology trials place robots in direct contact with patients, the contact behavior is set by machine-generated code, and no Federal law requires verification to come first. In a data pipeline a wrong number is caught downstream; in an embodied system the same error is a wrong motion no human can catch in time.

The solution: a new section 515D

The bill adds section 515D to the Federal Food, Drug, and Cosmetic Act (21 U.S.C. § 360e-5): no robot-patient interaction code may be generated or executed unless an automated verification, validation, and uncertainty quantification gate has first cleared, with the record documented and attested. A record made after the fact does not satisfy the section.

What it changes

Ten conforming amendments thread the rule through Title 21 so it is enforceable and consistent, from the device definition (§ 201(h)) through prohibited acts (§ 301), adulteration (§ 351), premarket review (§ 360e), and the State-law savings clause (§ 360k).

The assurance evidence

The requirement is feasible today, demonstrated on the record and reproducible from a fixed seed.

Figure placeholder (Images/evidence-to-law.png). The float, caption, and \label mirror the VVUQ-02 figures; replace the box with \includegraphics.

Figure 1: Figure 1. Evidence-to-law lineage placeholder: VVUQ-01/02 engineering proof feeding the section 515D rule.

Cost and the ask

Rulemaking only; no new program, expected PAYGO neutral. The ask is a House sponsor to introduce H. R. 9510 and refer it to Energy and Commerce.

Work	What it is	Releases	Headline result	DOI
VVUQ-01	method paper plus pipeline	v0.6.0	51/51 tests; fraction 1.0	xxxxxxx
VVUQ-02	ten-gate humanoid codebase	v1.2.0	172/172 tests; mean 93.56	xxxxxxx
VVUQ-03	standalone bill (H. R. 9510)	v1.4.0	verify-before-generate as statute	xxxxxxx

Table 1: Table 1. The supporting works at a glance (modeled on VVUQ-05 Table 1).

Rights and Permissions

This summary is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0). To view a copy, visit <https://creativecommons.org/licenses/by/4.0/>. Reproduced public-domain U.S. Government text is used under 17 U.S.C. § 105.

Cite This Summary

Kawchak K. One-Page Summary of H. R. 9510. Zenodo. 2026; 10.5281/zenodo.xxxxxxx.

Data Availability

Supporting records are openly available in the kevinkawchak/cancer-automated and kevinkawchak/physical-ai-oncology-trials GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

- [1] K. Kawchak, “VVUQ Processing Priority over Code Generation: Method and Pipeline.” <https://doi.org/10.5281/zenodo.20372501>, 2026.
<https://doi.org/10.5281/zenodo.20372501>.

-
- [2] K. Kawchak, “Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ.”
<https://doi.org/10.5281/zenodo.20421754>, 2026.
<https://doi.org/10.5281/zenodo.20421754>.
- [3] K. Kawchak, “Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510).” <https://doi.org/10.5281/zenodo.20535429>, 2026.
<https://doi.org/10.5281/zenodo.20535429>.
- [4] K. Kawchak, “National Physical AI Oncology Trial Platform.”
<https://doi.org/10.5281/zenodo.19244918>, 2026.
<https://doi.org/10.5281/zenodo.19244918>.
- [5] Creative Commons, “Attribution 4.0 International (CC BY 4.0).”
<https://creativecommons.org/licenses/by/4.0/>, 2013.
<https://creativecommons.org/licenses/by/4.0/>.
- [6] L. Lamport, *LaTeX: A Document Preparation System*. Addison-Wesley, 2nd ed., 1994.